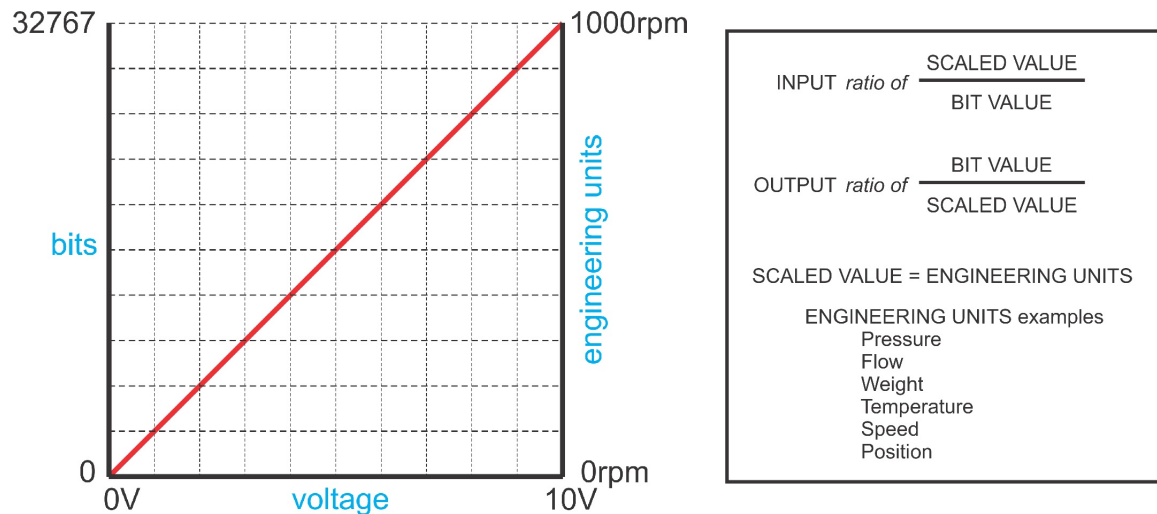

SCALING

For analog inputs and outputs, scaling is defining a ratio of the engineering units, the value range being measured, and the bit values produced by the electrical signal.



SCALING CALCULATIONS

bit min or max: minimum or maximum bit value of the module

scaled min or max: minimum or maximum value in engineering units, i.e. 0 psi. to 500 psi

scaled value: value in engineering units to determined from bit value

bit value: binary equivalent of scaled value

ANALOG INPUT (Bit value to Scaled Value)

Scaled value = (bit value x slope) + offset

Slope = (scaled max. – scaled min.) / (bit max. – bit min.)

Offset = scaled min. – (bit min. x slope)

ANALOG OUTPUT (Scaled value to Bit value)

Bit value = (scale value x slope) + offset

Slope = (bit max. – bit min.) / (scaled max. – scaled min.)

Offset = bit min. – (scaled min. x slope)

ANALOG INPUT EXAMPLES

Scaled Value: 0 to 1000 rpm

Bit Minimum and Maximum of Module: 0 to 32767

Bit Value from Analog Input: 16384

$$\text{Slope} = (1000 - 0) / (32767 - 0), \text{Slope} = 1000/32767 = 0.03051$$

$$\text{Offset} = 0 - (0 \times 0.031) = 0$$

$$\text{Scaled Value} = (16384 \times 0.03051) + 0 = 499.8758 \text{ rpm}$$

Scaled Value: 100° C to 500° C

Bit Minimum and Maximum of Module: 0 to 32767

Bit Value from Analog Input: 16384

$$\text{Slope} = (500 - 100) / (32767 - 0), \text{Slope} = 400/32767 = 0.01220$$

$$\text{Offset} = 100 - (0 \times 0.01120) = 100$$

$$\text{Scaled Value} = (16384 \times 0.01220) + 100 = 199.8848 + 100 = 299.8848 \text{ ° C}$$

ANALOG OUTPUT EXAMPLE

Scaled Value: 0 to 1000 rpm

Bit Minimum and Maximum of Module: 0 to 32767

Scaled Value to Analog Output: 500 rpm

$$\text{Slope} = (32767 - 0) / (1000 - 0), \text{Slope} = 32767/1000 = 32.767$$

$$\text{Offset} = 0 - (0 \times 0.031) = 0$$

$$\text{Scaled Value} = (500 \times 32.767) + 0 = 16,383.5 \text{ [16,384 bit value]}$$